

Optimisation Data Analyst Case Study

As part of your interview process with Adyen, you'll do a case study interview. We ask you to use Python, PySpark, or SQL for the analysis. We also ask you to share your presentation slides and your notebook(s) you use for the data analysis 24 hours before your interview.

This is based on anonymized merchant data for one merchant. We hope this case will give you a good sense of what an Optimisation Data Analyst's day-to-day role is like.

You'll be presenting your findings on the data and business recommendations to improve authorization rates for the anonymized merchant, defined as $\# \text{ of authorized transactions} / \# \text{ of total received transactions}$ and how to facilitate an experiment to prove the value add for your recommendation. During this case study, you'll be required to cover the following tasks:

1. Share your insights and clear recommendations to improve authorization rates, using the data provided. Keep in mind that your target audience will be executives from our commercial team, who rely on these insights for informed decision-making related to our customer partnerships.
2. Design a hypothesis test, with proper framework to outline how you could validate these recommendations through an experiment with statistical significance.
3. How would you approach scaling the insights from this case study across all merchants? Explain your thought process behind designing a scalable and sustainable solution, from analytical framework to technical implementation.

Your ability to connect your future work to the broader merchant landscape will be crucial.

Here's how the interview will be structured:

- 5 minutes: Introductions
- 20 minutes: Presentation
- 25 minutes: Q&A (Interviewers asking you questions. This can also be during your presentation, if you'd prefer questions sprinkled throughout instead of all at the end.)
- 10 minutes: Your questions to the interviewers if any

Below is a short description of the variables in the data:

- psp_reference: Adyen's internal index number for a transaction.
- bin: Bank Identification Number, the first 6 digits you see on a card.
- scheme: card scheme, whether it's Visa, MasterCard, or some other network.
- issuername: name of the bank that issued the card.
- shopper_interaction: channel of communication. "Ecommerce" means a one-off online transaction, while "ContAuth" stands for recurring transactions.
- avs_data_supplied: if Address data is supplied for that transaction.
- cvc_data_supplied: if the CVC (card verification number, the 3 digit in the back of the card) is supplied.
- currency_code: transaction currency.
- amount: transaction amount in local currency.
- raw_acquirer_response: issuer bank's response code to a transaction.
- creation_date: date and time (CET)
- received: denotes a received transaction
- authorization: if a transaction is approved (1) or refused (0).

Link to resources:

- [Database for the Case Study in Google Drive](#)
- [A detailed explanation of the acquirer responses](#)
- [Payment Glossary](#)
- <https://www.adyen.com/>
- [Adyen Formula](#)

Good luck and enjoy!

Best,

Adyen Optimization Data Analytics Team